

Generation Interconnection System Impact Study Report

For Queue Project Q22-2057

Grand Ridge Battery Projects 500 kV

March 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2057. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Grand Ridge Battery Projects 500 kV substation in the Illinois area. The request was received on April 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 711 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,245,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TIDD / LERDO 500.0 kV	EL PASO / RAVENNA 500.0 kV	104.9	2,594	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TIDD / LERDO 500.0 kV	Reconductor line	\$26,733,000	8% (\$2,245,000)
	Total allocated to project		\$2,245,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.